

EXPERIMENT 3

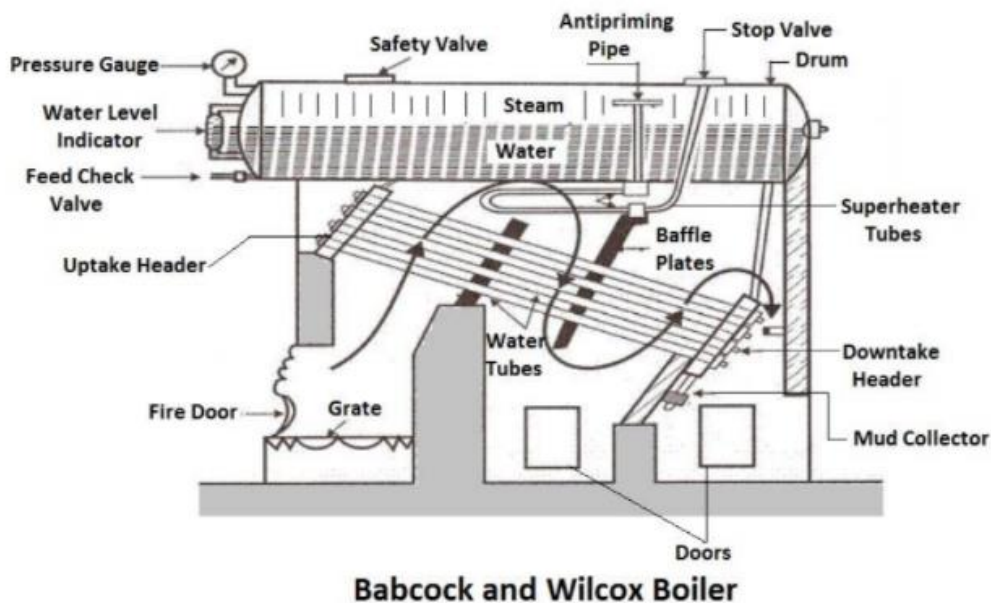
OBJECTIVE: To study Babcock Wilcox Boiler.

APPARATUS and MATERIALS USED: Cut section model of Babcock Wilcox Boiler.

THEORY:

Babcock And Wilcox Boiler is a stationary water tube boiler. It consists of the steam-water drum, a short tube connects the steam water drum with the uptake header and down header. A mud box is equipped with each down space header to collect the mud.

There is a slow-moving chain on which coal is fed from the hopper. The firebricks baffle wall deflected the hot gases and help the hot gases to leave the chimney. The dampers are used to regulate the draught and are operated by a chain and a pulley.



Parts of Babcock and Wilcox Boiler:

Following are the important parts of Babcock and Wilcox Boiler.

1. Water drum
2. Down take header
3. Uptake header
4. Water tubes
5. Baffle plates
6. Fire door
7. Grate
8. Mud box
9. Feed check valve
10. Furnace
11. Dampers

1. Water Drum

It is a horizontal axis drum containing water and steam. It is connected by a short tube with an uptake header or riser at the back end.

2. Downtake Header

This is present at the rear end of the boiler and joins the water tubes to the rear end of the drum. It collects the water from the drum.

3. Uptake Header

It is present at the front end of the boiler and is attached to the front end of the drum. It transmits the steam from water tubes to the drum.

4. Water Tubes

These are the tubes in which water passes and changes into steam. The water tubes (10 cm diameter) are inclined to the horizontal and connect the uptake header to the down take header. Due to its inclination, the water tubes are not completely filled with water, and water and steam are easily separated.

5. Baffle Plates

The Baffle plates are present within the water tubes. It causes hot gases to move upwards and downwards and again upwards before leaving the chimney. Baffle plates are provided to deflect hot flue gases.

6. Fire Door

It is applied to burn solid fuel in the furnace.

7. Grate

It is a platform, in which the solid fuel (coal) is burnt.

8. Mud Box

A mud box is provided with each down take header and the mud, that is settled down is removed.

9. Feed Check Valve

It is employed to fill water in the drum.

10. Furnace

The furnace is placed under the uptake header. In which the fuel is actually burnt.

11. Dampers

The dampers are operated by a chain that passes over a pulley to the front of a boiler to regulate the draught.

Apart from these , we have following mountings:

1. Water Level Indicator

It is an important fitting, which indicates the water level inside the boiler drum to an observer.

2. Pressure Gauge

A pressure gauge is used to measure the pressure of the steam inside the boiler drum.

3. Safety Valve

This a device attached to the steam chest for preventing explosions due to excessive internal pressure of steam.

Babcock Wilcox boilers also have a **Superheater** that is used to heat the steam to a superheated state. Superheated state of steam is basically completely dry steam whose temperature is above the saturation temperature of the steam for the same saturation pressure.

Working of Babcock and Wilcox Boiler

Coal is fed to the grate through the fire door and is burnt. The hot exhaust gases expand upward and pass across the left side of the water tubes. The baffles deflect the flue gases and hence the flue gases travel in a zig-zag manner over the water tubes and along with the superheater. The exhaust gases leave the atmosphere through the chimney.

The portion of the water tubes, which is just above the furnace, is heated comparatively to a higher temperature than the rest of it. Water rises into the drum through the uptake header. Here the steam and water are distributed in the drum. Steam, which is lighter, is being collected in the upper part of the drum. The water from the drum comes down through the down header into the water tubes.

Continuous circulation of water from the drum to the water tubes and vice versa is thus maintained. The circulation of water is maintained by convective currents and is known as "nature circulation". Steam is taken from the steam space through tubes to the superheater. Steam is superheated in the superheater.

The boiler is fitted with the necessary mountings for safe operation. The water level indicator and pressure gauge are attached to the boiler at its left end. The stop valve and steam safety valve are mounted on the upper side of the drum. Blow-off cock is provided for the periodical removal of mud and sediments collected in the mud box.

Advantages of Babcock and Wilcox Boiler

Following are the advantages of Babcock and Wilcox boiler:

1. This boiler produces steam up to 2000 to 40000 kg/hr.
2. It takes less space as compared to other boilers.
3. Boiler tubes can be easily replaced.
4. It is the only boiler that is used to produce a huge quantity of heat in power plants.
5. In this boiler, the draught loss is low.
6. It can be easily repaired and cleaned.
7. High overall efficiency.

Disadvantages of Babcock and Wilcox Boiler

Following are the disadvantages of Babcock and Wilcox boiler:

1. Large maintenance cost.
2. It is not suitable for impure and sedimentary water. In the case of impure and sedimentary water, the scale may accumulate in the tubes and this causes the tubes to overheat and burst. The water treatment is necessary before feeding in a boiler.
3. A continuous feedwater supply is needed to work. If continuous water is not supplied even for a small duration of time, in this case, the boiler is overheated. The water level must be strictly observed during the process of the Babcock and Wilcox boilers.

Application:

The Babcock and Wilcox boiler are generally used to produce high-pressure steam in power generation industries. The high-pressure steam so generated is used to produce electricity.

OBSERVATIONS:

[Students need to paste an actual photos of the model and label the parts accordingly in a fresh Page]